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MATHEMATICS

STANDARD SIX

Terminal exam - May 2026
Answer Key and Worked Solutions

Subject	Mathematics
Code	Not printed in source
Level	Standard Six
Year	2026
Exam Board / Issuer	Arusha region
Exam	Terminal exam - May
Type	Answer key and worked solutions
Questions	33 numbered questions/subparts (8 main questions)

Try each question before checking the solution. Follow the steps, compare your working, and practise the parts you find difficult.

Source paper: Mathematics-S6-2026-Arusha-region-May-Terminal-exam.pdf

Notes about the source paper

Items with missing information or printing errors are explained in their solutions. A model response is an example, rather than a prescribed official answer.

Section A: Mathematical operations

1(i). Calculate $9637 + 64387$.

Step 1: 1 place: $7 + 7 = 14$; write 4, carry 1.

Step 2: 10 place: $3 + 8 + \text{carry } 1 = 12$; write 2, carry 1.

Step 3: 100 place: $6 + 3 + \text{carry } 1 = 10$; write 0, carry 1.

Step 4: 1000 place: $9 + 4 + \text{carry } 1 = 14$; write 4, carry 1.

Step 5: 10000 place: $0 + 6 + \text{carry } 1 = 7$; write 7.

Step 6: $9637 + 64387 = 74024$.

Final Answer: 74024

1(ii). Calculate $14938 - 8750$.

Step 1: 1 place: $8 - 0 = 8$.

Step 2: 10 place: regroup 1 from the next place; $13 - 5 = 8$.

Step 3: 100 place: $8 - 7 = 1$.

Step 4: 1000 place: regroup 1 from the next place; $14 - 8 = 6$.

Step 5: 10000 place: $0 - 0 = 0$.

Step 6: $14938 - 8750 = 6188$; check $6188 + 8750 = 14938$.

Final Answer: 6188

1(iii). Calculate 118×34 .

Step 1: Split into place values: $100 \times 34 = 3400$; $10 \times 34 = 340$; $8 \times 34 = 272$.

Step 2: Add the partial products: 4012.

Final Answer: 4012

1(iv). Calculate $1500 \div 250$.

Step 1: $1500 \div 250$: write 6; $6 \times 250 = 1500$; $1500 - 1500 = 0$.

Step 2: $250 \times 6 = 1500$.

Step 3: $1500 \div 250 = 6$.

Final Answer: 6

1(v). Calculate $9 \frac{1}{6} - 8 \frac{2}{3}$.

Step 1: $9 \frac{1}{6} = \frac{55}{6}$ and $8 \frac{2}{3} = \frac{26}{3} = \frac{52}{6}$.

Step 2: $\frac{55}{6} - \frac{52}{6} = \frac{3}{6} = \frac{1}{2}$.

Final Answer: $\frac{1}{2}$

1(vi). Calculate $2 \frac{3}{5} + 1 \frac{2}{3}$.

Step 1: $2 \frac{3}{5} = \frac{13}{5}$ and $1 \frac{2}{3} = \frac{5}{3}$.

Step 2: Use denominator 15: $\frac{13}{5} = \frac{39}{15}$ and $\frac{5}{3} = \frac{25}{15}$.

Step 3: $\frac{39}{15} + \frac{25}{15} = \frac{64}{15} = 4 \frac{4}{15}$.

Final Answer: $4 \frac{4}{15}$

1(vii). Calculate $-(44) - (-15)$.

Step 1: Subtracting a negative number means adding the corresponding positive number.

Step 2: $-44 - (-15) = -44 + 15 = -29$.

Final Answer: -29

1(viii). Calculate 0.045×0.03 .

Step 1: Without the decimal points, $45 \times 3 = 135$.

Step 2: There are $3 + 2 = 5$ decimal places in the factors.

Step 3: Restore five decimal places: 0.00135.

Final Answer: 0.00135

1(ix). Calculate $3.86 + 17.9$.

Step 1: Write 17.9 as 17.90 and align the decimal points.

Step 2: Hundredths: $6 + 0 = 6$; tenths: $8 + 9 = 17$, write 7 and carry 1 to the units.

Step 3: Whole-number part: $3 + 17 + 1 = 21$.

Step 4: $3.86 + 17.90 = 21.76$.

Final Answer: 21.76

1(x). Calculate $1 \frac{1}{2} \div 2 \frac{1}{3}$.

Step 1: Convert mixed numbers: $1 \frac{1}{2} = \frac{3}{2}$ and $2 \frac{1}{3} = \frac{7}{3}$.

Step 2: Divide by multiplying by the reciprocal: $(\frac{3}{2}) \times (\frac{3}{7})$.

Step 3: $3 \times 3 = 9$ and $2 \times 7 = 14$, giving $\frac{9}{14}$.

Final Answer: $\frac{9}{14}$

Section Summary: Align place values and decimal points. Convert mixed numbers to improper fractions; use a common denominator for addition and subtraction and the reciprocal for division.

Section B: Word problems

2(i). Convert 1.09 into a percentage.

Step 1: Multiply by 100 to express the number per hundred.

Step 2: $1.09 \times 100 = 109$.

Final Answer: 109%

2(ii). List all odd numbers between 70 and 80 that are also square numbers.

Step 1: The consecutive nearby integer squares are $8^2 = 64$ and $9^2 = 81$.

Step 2: No integer square lies between 70 and 80, so no odd square qualifies.

Final Answer: None

2(iii). Write the smallest number formed from the digits 6, 4, 9, 0, 2, 7.

Step 1: Using the listed digits once each, choose the smallest nonzero digit 2 first.

Step 2: Arrange the remaining digits in ascending order: 0, 4, 6, 7, 9.

Step 3: The number is 204679.

Final Answer: 204679

2(iv). Gwakisa was 10 years old when Tanganyika became independent in 1961. Find his age in the paper year 2026 and write it in Roman numerals.

Step 1: Years elapsed = $2026 - 1961 = 65$.

Step 2: Age = $10 + 65 = 75$ years.

Step 3: $75 = 50 + 20 + 5 = L + XX + V = LXXV$.

Final Answer: LXXV (75 years)

2(v). A football match began at 1830 hours. Write this time in the 12-hour format.

Step 1: 18 is greater than 12, so subtract 12: $18 - 12 = 6$.

Step 2: Keep the 30 minutes and write p.m.

Final Answer: 6:30 p.m.

2(vi). Write MCDXLIX in Arabic numerals.

Step 1: $M = 1000$; $CD = 400$; $XL = 40$; $IX = 9$.

Step 2: $1000 + 400 + 40 + 9 = 1449$.

Final Answer: 1449

3(i). Write the missing number: 2, 3, 5, 7, __.

Step 1: The listed numbers are consecutive prime numbers.

Step 2: 8, 9 and 10 are composite; the next prime after 7 is 11.

Final Answer: 11

3(ii). Consolatha's car has mass 1500.008 kg. Round the mass to two decimal places.

Step 1: The hundredths digit is 0; the next digit, in the thousandths place, is 8.

Step 2: Since 8 is at least 5, raise the hundredths digit to 1.

Final Answer: 1500.01 kg

3(iii). Find the LCM of 24, 28 and 50.

Step 1: $24 = 2^3 \times 3$; $28 = 2^2 \times 7$; $50 = 2 \times 5^2$.

Step 2: Use the largest power of each prime factor: $\text{LCM} = 2^3 \times 3 \times 7 \times 5^2$.

Step 3: $8 \times 3 \times 7 \times 25 = 4200$.

Final Answer: 4200

4(i). 12 people cultivate a farm in 4 days. Four more people of equal ability join. How long will the same work take?

Step 1: Total work = $12 \times 4 = 48$ person-days.

Step 2: New number of workers = $12 + 4 = 16$.

Step 3: New time = $48/16 = 3$ days.

Final Answer: 3 days

4(ii). Simplify $(mn - n)/(n - mn)$.

Step 1: Factor the numerator: $mn - n = n(m - 1)$.

Step 2: Factor the denominator: $n - mn = n(1 - m) = -n(m - 1)$.

Step 3: Cancel the common nonzero factor $n(m - 1)$; the result is -1.

Step 4: The original expression is defined only when n is nonzero and m is not 1.

Final Answer: -1, provided $n \neq 0$ and $m \neq 1$.

4(iii). How many $1/4$ are there in 40?

Step 1: Number of quarters = $40/(1/4)$.

Step 2: Divide by a fraction by multiplying by its reciprocal: $40 \times 4 = 160$.

Final Answer: 160

5(i). Eight players share money and each receives Sh 42000. If three leave, how much would each remaining player receive from the same total?

Step 1: Total money = $8 \times 42000 = \text{Sh } 336000$.

Step 2: Remaining players = $8 - 3 = 5$.

Step 3: Each receives $336000/5 = \text{Sh } 67200$.

Final Answer: Sh 67200 per player

5(ii). 8 kg of coffee costs Sh 19200. Find the cost of 6 kg of the same coffee.

Step 1: Cost per kilogram = $19200/8 = \text{Sh } 2400$.

Step 2: Cost of 6 kg = $2400 \times 6 = \text{Sh } 14400$.

Final Answer: Sh 14400

5(iii). Masharubu buys 6 kg of meat at Sh 1200 per kg, 6 soap bars at Sh 440 each and 4 kg of sugar at Sh 600 per kg. Find the total bill.

Step 1: Meat: $6 \times 1200 = \text{Sh } 7200$.

Step 2: Soap: $6 \times 440 = \text{Sh } 2640$.

Step 3: Sugar: $4 \times 600 = \text{Sh } 2400$.

Step 4: Total = $7200 + 2640 + 2400 = \text{Sh } 12240$.

Final Answer: Sh 12240

6(i). Martha is 3 years older than Semen. Their ages total 37 years. Find Martha's age.

Step 1: Let Semen's age be a years, so Martha's age is $a + 3$.

Step 2: $a + (a + 3) = 37$, so $2a = 37 - 3 = 34$.

Step 3: $a = 34/2 = 17$; Martha is $17 + 3 = 20$.

Step 4: Check: $17 + 20 = 37$ and $20 - 17 = 3$.

Final Answer: 20 years

6(ii). Divide 8 hours 03 minutes by 7.

Step 1: Convert the duration to minutes: $8 \times 60 + 3 = 483$ minutes.

Step 2: $483/7 = 69$ minutes.

Step 3: 69 minutes = 1 hour 9 minutes.

Final Answer: 1 hour 9 minutes

6(iii). Find K if $2K + 8 = 3K - 12$.

Step 1: Subtract 2 K from both sides: $8 = K - 12$.

Step 2: Add 12 to both sides: $K = 8 + 12 = 20$.

Step 3: Check: $2 \times 20 + 8 = 48$ and $3 \times 20 - 12 = 48$.

Final Answer: $K = 20$

Section Summary: Choose the operation from the question, convert to consistent units, and check the result. LCM uses the highest prime powers; GCF uses only factors shared by all numbers.

Section C: Geometry and statistics

7(i)(a). Mr. Nzunda's salary pie chart shows house rent with a right-angle mark (90°), food 125° and school fees in the remaining sector. Find the percentage spent on house rent.

Step 1: Rent takes 90° out of 360° .

Step 2: Rent fraction = $90/360 = 1/4$.

Step 3: Percentage = $(1/4) \times 100 = 25$.

Final Answer: 25%

7(i)(b). In the same pie chart, Mr. Nzunda spends Sh 2400 on the 90° house-rent sector. Find his total monthly salary.

Step 1: Rent is $90/360 = 1/4$ of the salary.

Step 2: The whole salary is therefore $2400 \times 4 = \text{Sh } 9600$.

Step 3: Check: $9600 \times 1/4 = 2400$.

Final Answer: Sh 9600

8(i). A cylinder is open at both ends, has diameter 14 cm and length 20 cm. Find its surface area.

Step 1: Open ends have no circular bases, so use only the curved surface.

Step 2: Curved area = circumference x length = $\pi \times \text{diameter} \times \text{length}$.

Step 3: Area = $\pi \times 14 \times 20 = 280 \pi \text{ cm}^2$.

Step 4: Using $\pi = 22/7$ gives $(22/7) \times 14 \times 20 = 880 \text{ cm}^2$.

Final Answer: $280 \pi \text{ cm}^2$ (880 cm^2 using $\pi = 22/7$)

8(ii). Angles 60° , $3x$ and $2x$ are adjacent on a straight line. Find x .

Step 1: Angles on a straight line total 180° : $60 + 3x + 2x = 180$.

Step 2: $5x = 180 - 60 = 120$.

Step 3: $x = 120/5 = 24^\circ$.

Step 4: Check: $60 + 3 \times 24 + 2 \times 24 = 180$.

Final Answer: $x = 24^\circ$

8(iii). Mpemba Primary School's cuboid water tank measures 100 cm by 20 cm by 10 cm. Find its full capacity in litres, using $1000 \text{ cm}^3 = 1 \text{ litre}$.

Step 1: Volume = length x width x height = $100 \times 20 \times 10 = 20000 \text{ cm}^3$.

Step 2: Convert to litres: $20000/1000 = 20$.

Final Answer: 20 litres

Section Summary: Use marked measurements and angles. Circle sectors are fractions of 360° . Right triangles obey $a^2 + b^2 = c^2$; cylinder volume is $\pi \times r^2 \times h$. Read graph coordinates as (x, y) .

Source Checks

Darasa Huru - Arusha Region Standard Six Terminal Exams May 2026